



COMSATS University Islamabad, Lahore Campus

Department of Computer Sciences

Terminal Lab – Semester Spring 2026

Course Title	Software Testing			Course Code	CSE482	Credit Hours:	3
Course Instructor	Ms. Yella Mehroze			Program Name	BSE		
Semester	8 th	Batch:	FA22-BSE	Section	A	Type	Theory
Total Marks	50	Obtained Marks:		Date	May 09, 2026		
Student's Name	UMAIR ALI			Reg. No	FA22-BSE-137		

Question1:

[Marks 20]

[CLO <5>]: Create test plan document for a medium size software system in a team environment.

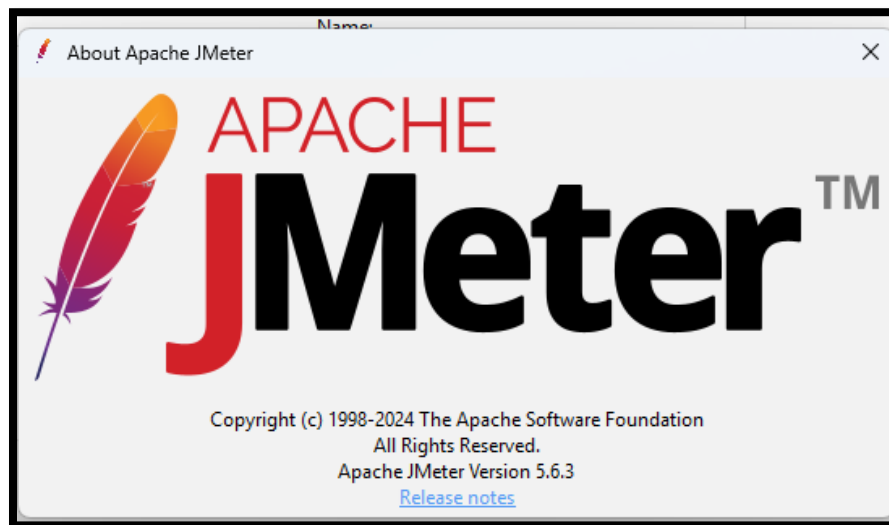
Bloom Taxonomy Level<Create>

Part A - Choose the Apache JMeter to perform performance Testing.

INSTALLATION — DO THIS FIRST

1. Install Java JDK 11+ → <https://adoptium.net>
2. Download JMeter 5.6+ → https://jmeter.apache.org/download_jmeter.cgi
3. Extract to a folder → e.g. C:\JMeter or /opt/jmeter
4. Launch JMeter GUI → Windows: bin\jmeter.bat | Linux/Mac: ./bin/jmeter.sh
5. Confirm installation → Help menu → About Apache JMeter → must show version 5.6+

```
[ SWEET ]—[ SUNDAY 17/05/2026 - 5:02:53 PM ]—[ C:\Portable\AST-Terminal ]
[ UMAIR •• PowerShell ] :] java --version
openjdk 25.0.3 2026-04-21 LTS
OpenJDK Runtime Environment Microsoft-13877124 (build 25.0.3+9-LTS)
OpenJDK 64-Bit Server VM Microsoft-13877124 (build 25.0.3+9-LTS, mixed mode, sharing)
```



TASK 1 Foundations — GUI, HTTP Requests & Assertions

Covers: Installation · Thread Groups · HTTP Methods · Assertions

This task builds the complete foundation of JMeter. By the end you will understand the GUI, be able to send all types of HTTP requests, and validate responses using assertions.

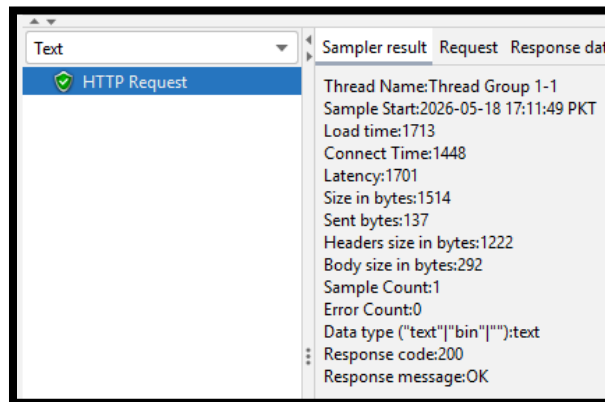
Part A — GUI Orientation & First Request

After installing JMeter, explore the interface and send your first HTTP request to a live API.

1. Launch JMeter and identify these 5 components: Test Plan, Menu Bar, Toolbar, Tree Panel (left), Detail Panel (right)
2. Right-click Test Plan → Add → Threads → Thread Group. Set: 1 user, Ramp-Up 1s, Loop Count 1
3. Right-click Thread Group → Add → Sampler → HTTP Request. Configure as follows:

```
Protocol:      https
Server Name:   jsonplaceholder.typicode.com
Method:        GET
Path:          /posts/1
```

4. Add two listeners: View Results Tree and Summary Report
5. Click the green Run button. Confirm a GREEN result. Note the response time and response code



Summary Report

Name:

Summary Report

Comments:

Write results to file / Read from file

Filename

Browse...

Log/Display Only:

☐ Errors

☐ Successes

Configure

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec	Avg. Bytes
HTTP Request	1	1713	1713	1713	0.00	0.00%	35.0/min	0.86	0.08	1514.0
TOTAL	1	1713	1713	1713	0.0	0.00%	35.0/min	0.86	0.08	1514.0

Response Time: 0.86/sec received; 0.08/sec sent
Response Code: 200

B — All HTTP Methods

Extend your test plan to cover all four HTTP methods used in modern REST web APIs.

6. Add 3 more HTTP Request samplers to the same Thread Group for POST, PUT, and DELETE:

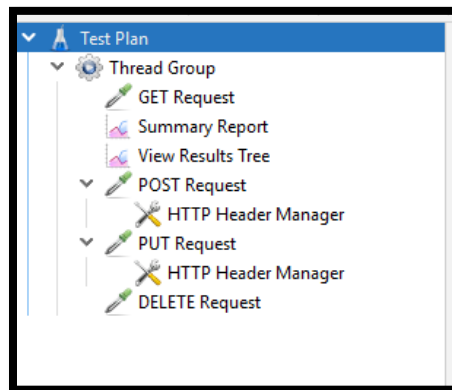
```
POST    /posts
  Body: {"title":"Test Post","body":"Hello World","userId":1}

PUT     /posts/1
  Body: {"id":1,"title":"Updated Title","body":"Updated Body","userId":1}

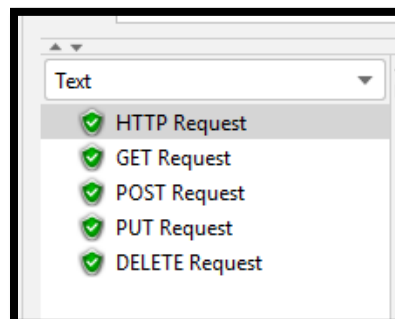
DELETE /posts/1
(no body needed)
```

For POST and PUT — add HTTP Header Manager to each sampler:
 Right-click sampler → Add → Config Element → HTTP Header Manager
 Name: Content-Type Value: application/json

7. Run all 4 samplers. Expected response codes: GET = 200, POST = 201, PUT = 200, DELETE = 200



Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec	Avg. Bytes
HTTP Request	1	1713	1713	1713	0.00	0.00%	35.0/min	0.86	0.08	1514.0
GET Request	1	648	648	648	0.00	0.00%	1.5/sec	2.28	0.21	1515.0
POST Request	1	605	605	605	0.00	0.00%	1.7/sec	2.26	0.39	1399.0
PUT Request	1	1190	1190	1190	0.00	0.00%	50.4/min	1.05	0.21	1279.0
DELETE Request	1	611	611	611	0.00	0.00%	1.6/sec	1.90	0.36	1188.0
TOTAL	5	953	605	1713	439.29	0.00%	5.1/hour	0.00	0.00	1379.0



GET Request Code: 200
 PUT Request Code: 200
 POST Request Code: 201
 DELETE Request Code: 200

✓ GET Request	Sample Start:2026-05-18 18:10:37 PKT
✓ POST Request	Load time:648
✓ PUT Request	Connect Time:459
✓ DELETE Request	Latency:647
	Size in bytes:1515
	Sent bytes:137
	Headers size in bytes:1223
	Body size in bytes:292
	Sample Count:1
	Error Count:0
	Data type ("text" "bin" ""):text
	Response code:200
	Response message:OK

GET

✓ PUT Request	Connect Time:0
✓ DELETE Request	Latency:1190
	Size in bytes:1279
	Sent bytes:254
	Headers size in bytes:1197
	Body size in bytes:82
	Sample Count:1
	Error Count:0
	Data type ("text" "bin" ""):text
	Response code:200
	Response message:OK

PUT

✓ POST Request	Load time:605
✓ PUT Request	Connect Time:0
✓ DELETE Request	Latency:605
	Size in bytes:1399
	Sent bytes:241
	Headers size in bytes:1320
	Body size in bytes:79
	Sample Count:1
	Error Count:0
	Data type ("text" "bin" ""):text
	Response code:201
	Response message:Created

POST

✓ DELETE Request	Latency:011
	Size in bytes:1188
	Sent bytes:223
	Headers size in bytes:1186
	Body size in bytes:2
	Sample Count:1
	Error Count:0
	Data type ("text" "bin" ""):text
	Response code:200
	Response message:OK

DELETE

Part C — Assertions

Add three types of assertions to the GET /posts/1 request to automatically validate the server response.

1. Response Code Assertion — verifies the HTTP status code is 200
2. Response Body Assertion — verifies the response body contains the word 'userId'
3. JSON Assertion — verifies the JSON field \$.userId equals the value 1

Assertion 1 — Response Code:

Right-click GET sampler → Add → Assertions → Response Assertion
Field to Test: Response Code Pattern: Equals Value: 200

Assertion 2 — Response Body Contains Text:

Add another Response Assertion
Field to Test: Response Body Pattern: Contains Value: userId

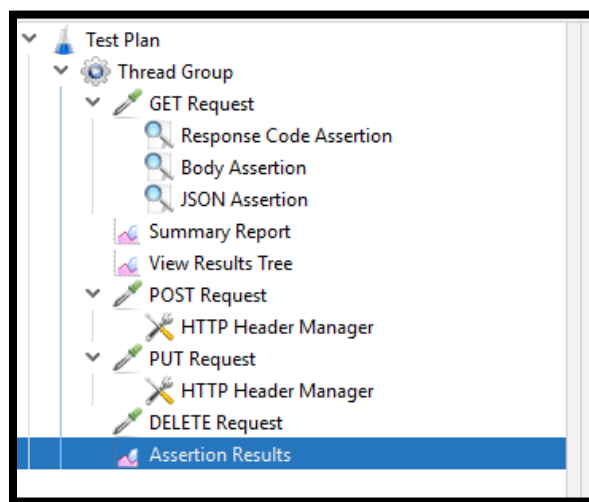
Assertion 3 — JSON Path Assertion:

Right-click GET sampler → Add → Assertions → JSON Assertion
JSON Path: \$.userId Expected Value: 1 Check 'Additionally assert value'

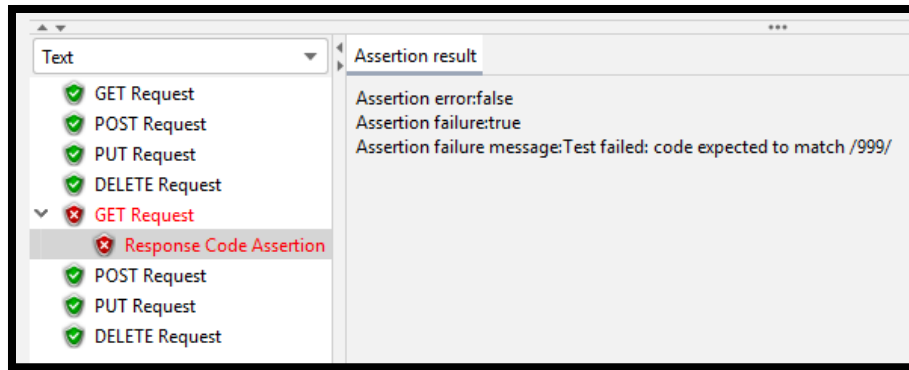
4. Intentional failure test: Change the Response Code assertion value to 999, run, and observe the RED failure. Then restore it back to 200 and confirm GREEN again

Task 1 — Deliverables:

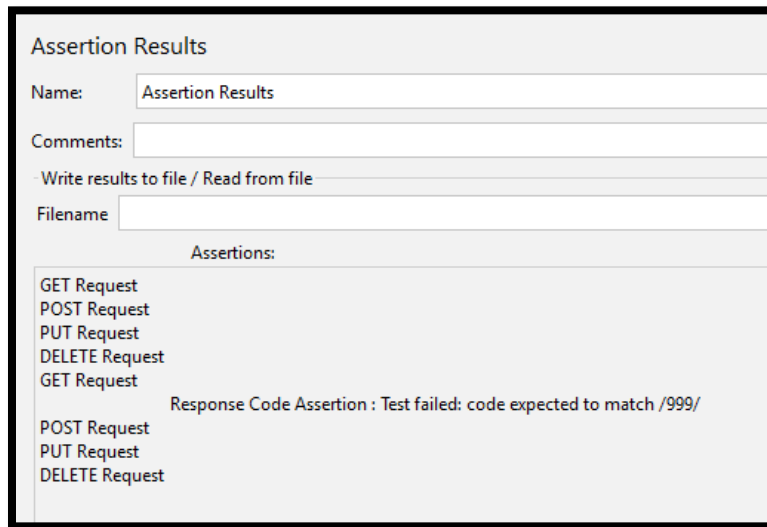
[]	Labelled screenshot of JMeter GUI identifying all 5 components
[]	Screenshot of View Results Tree showing the GET request passing (GREEN)
[]	Screenshot of all 4 HTTP methods in View Results Tree — all GREEN
[]	Screenshot of all 3 assertions passing on the GET request
[]	Screenshot of the intentional assertion failure showing RED result with error message



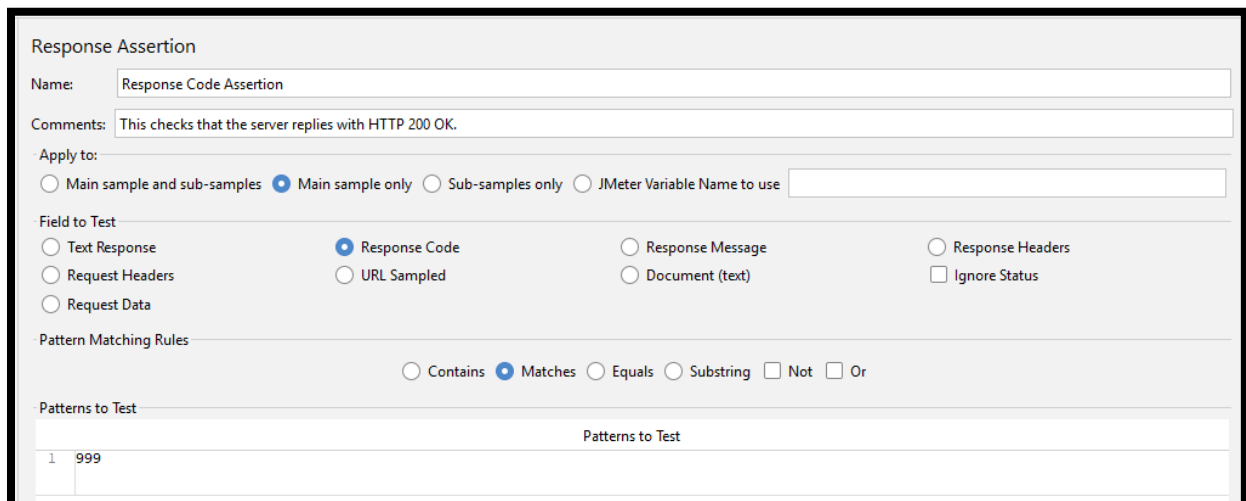
TREE



Success and failure test by changing response code to 999.



failure test in assertion results matching code 999



Response Code Assertion

Response Assertion

Name:

Comments:

Apply to:

☐ Main sample and sub-samples ☒ Main sample only ☐ Sub-samples only ☐ JMeter Variable Name to use

Field to Test

☒ Text Response ☐ Response Code ☐ Response Message ☐ Response Headers

☐ Request Headers ☐ URL Sampled ☐ Document (text) ☐ Ignore Status

☐ Request Data

Pattern Matching Rules

☒ Contains ☐ Matches ☐ Equals ☐ Substring ☐ Not ☐ Or

Patterns to Test

	Patterns to Test
1	userld

Body Assertion

JSON Assertion

Name:

Comments:

Assert JSON Path exists:

Additionally assert value ☒

Match as regular expression ☒

Expected Value:

	Expected Value
1	1

Expect null ☐

Invert assertion (will fail if above conditions met) ☐

JSON Assertion

TASK 2 Performance Testing — Load, Stress & Data-Driven

Covers: Load Test · Service Level Agreement Evaluation · Stress Test · CSV Parameterization · Correlation

This is the core of JMeter — simulating real users under load. You will run a formal load test against defined performance targets, find the system's breaking point with a stress test, and use CSV files to send different data for each virtual user.

Part A — Load Test with Service Level Agreement Evaluation

A Service Level Agreement (Service Level Agreement) defines the minimum performance standards a system must meet. You will run a load test and check whether the application meets all four targets below.

PERFORMANCE TARGETS — Service Level Agreement

Average Response Time → must be UNDER 2000 milliseconds

90th Percentile → 90% of all requests must complete in UNDER 3000 milliseconds

Error Rate → must be UNDER 1% of total requests

Throughput → server must handle at LEAST 5 requests per second

1. Target endpoint: GET <https://jsonplaceholder.typicode.com/posts>
2. Configure Thread Group:

```
Number of Threads (users): 50
Ramp-Up Period: 30 seconds
Loop Count: 10
Scheduler: checked Duration: 120 seconds (run for 2 minutes)
```

3. Add these listeners: Aggregate Report, Response Time Graph, Summary Report
4. Run the full 2-minute test. Record all 4 metrics from the Aggregate Report
5. Write a PASS or FAIL verdict for each metric against the Service Level Agreement targets above

Thread Group

Name: Load Test - 50 Users

Comments:

Action to be taken after a Sampler error

☒ Continue ☐ Start Next Thread Loop ☐ Stop Thread ☐ Stop Test ☐ Stop Test Now

Thread Properties

Number of Threads (users): 50

Ramp-up period (seconds): 60

Loop Count: ☒ Infinite

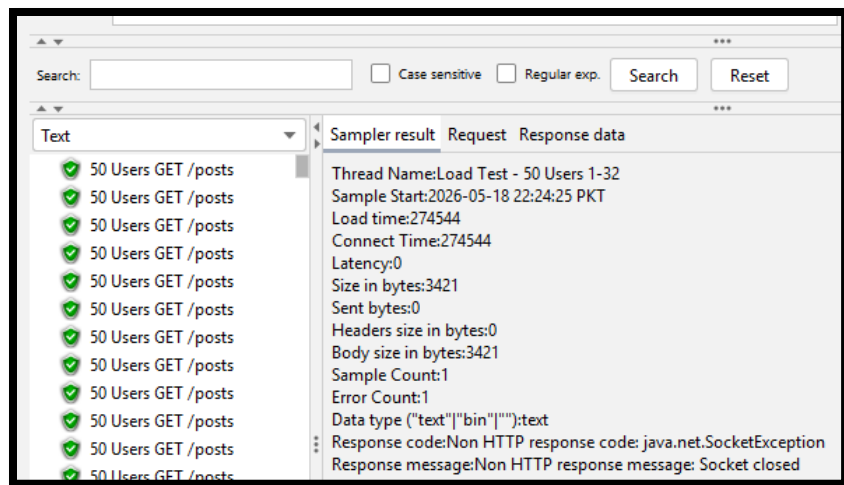
☒ Same user on each iteration

☐ Delay Thread creation until needed

☒ Specify Thread lifetime

Duration (seconds): 120

Startup delay (seconds): 0



Summary Report

Name:

Summary Report

Comments:

Write results to file / Read from file

Filename

Browse...

Log/Display Only:

☐ Errors

☐ Successes

Configure

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/ ...	Sent KB/sec	Avg. Bytes
50 Users GET ...	2501	1913	160	274544	5853.81	0.04%	8.0/sec	225.20	1.06	28727.9
TOTAL	2501	1913	160	274544	5853.81	0.04%	8.0/sec	225.20	1.06	28727.9

Aggregate Report

Name:

Aggregate Report

Comments:

Write results to file / Read from file

Filename

Browse...

Log/Display Only:

☐ Errors

☐ Successes

Configure

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughput	Received K...	Sent KB/sec
50 Users G...	2501	1913	1384	3530	4273	10795	160	274544	0.04%	8.0/sec	225.20	1.06
TOTAL	2501	1913	1384	3530	4273	10795	160	274544	0.04%	8.0/sec	225.20	1.06

Metric	SLA Target	Pass/Fail?
Average Response Time	< 2000ms	✓
Error Rate	< 1%	✓
Throughput	As high as possible	8.0/sec
90th Percentile	< 3000ms	✗

Part B — Stress Test: Find the Breaking Point

SERVICE LEVEL AGREEMENT vs STRESS TEST

Load Test: Verify system handles EXPECTED traffic within the Service Level Agreement targets

Stress Test: Push BEYOND normal limits to find WHEN the system starts failing

The user count where error rate first exceeds 5% is called the breaking point

- Run the same endpoint 4 times with increasing users. Stop when error rate exceeds 5%:

```
Run 1: 25 users, ramp-up 10s, loop 10 → record error %
Run 2: 75 users, ramp-up 15s, loop 10 → record error %
Run 3: 150 users, ramp-up 25s, loop 10 → record error %
Run 4: 300 users, ramp-up 40s, loop 10 → record error %
```

- Identify the breaking point — at how many users did error rate first exceed 5%?

Run	Users	Avg Response (ms)	90% Line (ms)	Error %	Throughput
1	100	713	708	0.00%	1.7/sec
2	200	656	756	0.98%	59.9/min
3	300	626	845	0.66%	1.3/sec
4	400	588	661	0.81%	1.7/sec

Part C — CSV Data-Driven Testing & Correlation

Real tests send different data for each virtual user instead of the same hardcoded value. This part covers CSV parameterization and passing data between two chained requests (called correlation).

8. Create a file named `test_data.csv` with this content:

```
post_id,expected_code
1,200
10,200
50,200
100,200
999,404
```

9. Add CSV Data Set Config to Thread Group (5 users, loop 1):

```
Right-click Thread Group → Add → Config Element → CSV Data Set Config
Filename:          full path to your test_data.csv
Variable Names:    post_id,expected_code
Delimiter:         ,
Stop thread on EOF: True
```

10. Set HTTP Request path to `/posts/${post_id}` and add Response Assertion using `${expected_code}`

11. Add a JSON Extractor to the HTTP Request to extract the returned post ID:

```
Right-click HTTP Request → Add → Post Processors → JSON Extractor
Variable name:    extracted_id
JSON Path:        $.id
```

Add a second HTTP Request using the extracted value:

Path: `/posts/${extracted_id}/comments`

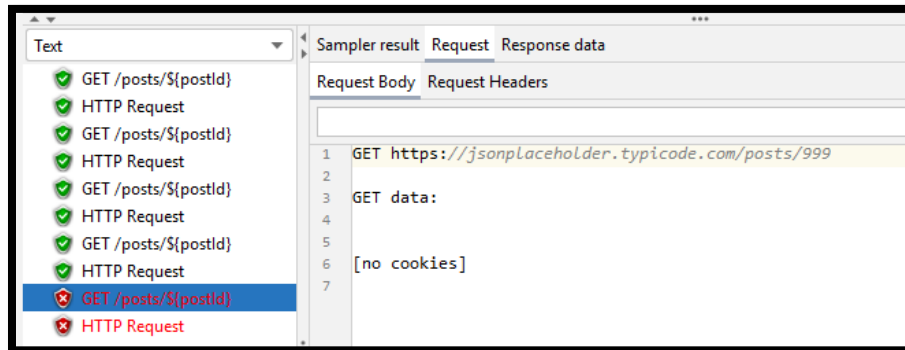
Verify in View Results Tree that Request 2 URL contains the correct extracted ID

Task 2 — Deliverables:

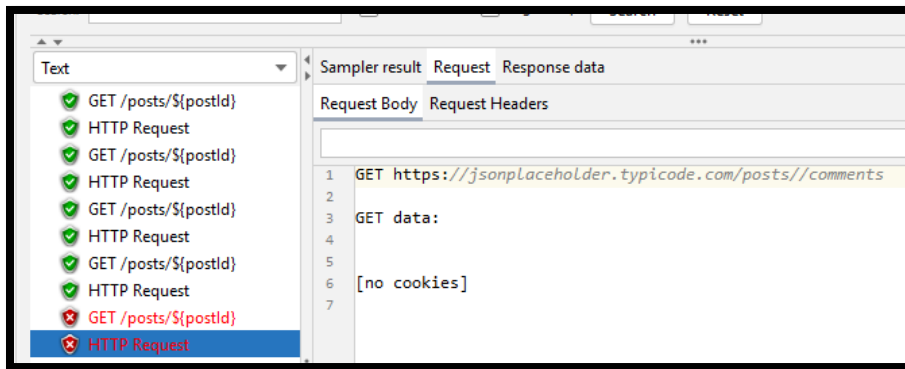
[]	Screenshot of Aggregate Report from the 50-user load test
[]	Service Level Agreement table in your report: metric measured value target PASS or FAIL
[]	Screenshot of Response Time Graph showing the full 2-minute run
[]	Stress test table: all 4 runs with user count and error percentage
[]	Written answer: What was the breaking point and at how many users did it occur?
[]	Screenshot of CSV Data Set Config settings
[]	Screenshot of View Results Tree showing 5 users hitting different endpoints
[]	Screenshot of View Results Tree showing Request 2 URL using the extracted ID
[]	Written answer: What is correlation and when is it needed?

```
{
  "userId": 1,
  "id": 1,
  "title": "sunt aut facere repellat provident occaecati excepturi optio reprehenderit",
  "body": "quia et suscipit\nsuscipit recusandae consequuntur expedita et cum\nreprehenderit molestiae ut ut quas tota\nrem aut distinctio\nvoluptates doloribus"
}
```

First User Request Response



Error due to post 999 not exist
999 id taken from JSON.



When no post 999 exist not comment found

5 Get Requests Show 5 Different Users Doing /Get Requests.

TASK 3 Professional JMeter — CI/CD Pipeline Integration

· GitHub Actions Automation

This task covers professional-grade JMeter usage as practiced in real software teams. You will run your test from the command line, generate a full HTML performance dashboard, and set up an automated pipeline on GitHub so your test runs every time code is pushed.

Non-GUI Reports & GitHub Actions CI/CD

IMPORTANT RULE FOR REAL PROJECTS

Never run actual load tests using the JMeter GUI — it uses extra CPU and memory which distorts results.

Always run performance tests from the command line (called non-GUI or headless mode).

Rule to remember: Use the GUI only to BUILD the test plan. Always RUN from the command line.

Step 1 — Run JMeter from the Command Line

12. Save your Task 2 load test as `load_test.jmx`

13. Open a terminal or command prompt and run the following command:

```
# Windows:
cd C:\apache-jmeter-5.6\bin
jmeter -n -t C:\path\load_test.jmx -l C:\results\results.jtl -e -o C:\reports\
```

```
# Linux / Mac:
cd /opt/jmeter/bin
./jmeter.sh -n -t ~/load_test.jmx -l ~/results/results.jtl -e -o ~/reports/
```

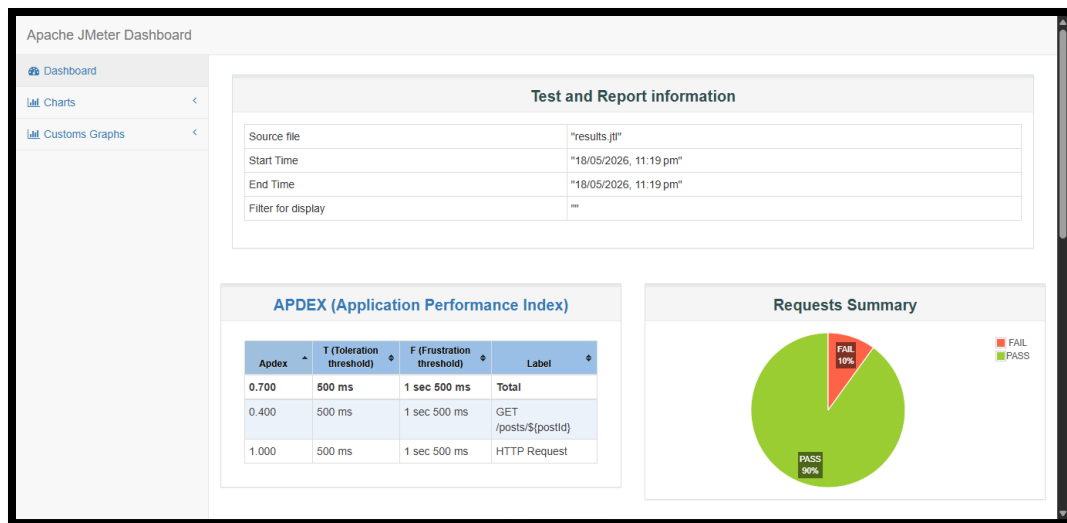
What each flag means:

- n = Non-GUI (headless) mode
- t = Path to your .jmx test plan file
- l = Path to save the raw results (.jtl file)
- e = Generate HTML report after test finishes
- o = Output folder for HTML report (must be empty before running)

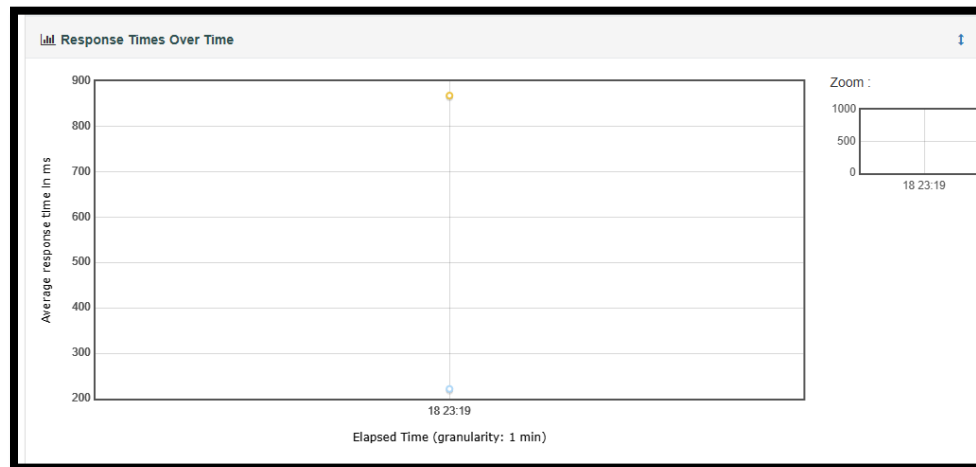
14. Open the generated report: browse to your reports folder and open `index.html` in a browser

15. Explore these 3 sections of the HTML dashboard and describe each one in your report:

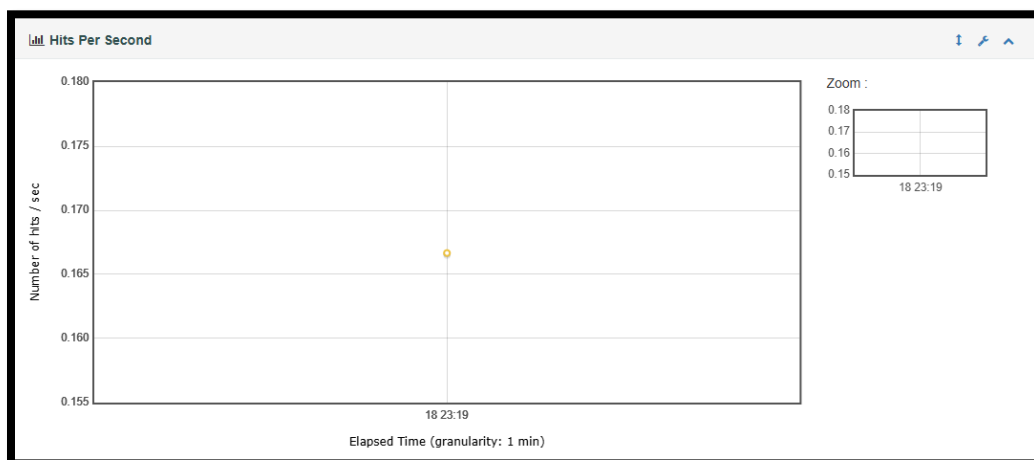
- Response Time Over Time chart — how did response time change during the test?
- Throughput Over Time chart — how many requests per second at each moment?
- Response Time Percentiles — what do the P90 and P99 values mean?



REPORT



RESPONSE TIME



THROUGHPUT

Step 2 — CI/CD Pipeline with GitHub Actions

Automate your JMeter test so it runs automatically every time code is pushed to GitHub. This is exactly how real Quality Assurance teams catch performance problems before they reach users.

16. Create a free GitHub account at github.com if you do not already have one
17. Create a new public repository named `jmeter-semester-project`
18. Upload your `load_test.jmx` file into a folder called `tests/` inside the repository
19. Create a new file at this path in the repository: `.github/workflows/jmeter.yml`
20. Add the following content to the workflow file:

```
name: JMeter Performance Tests

on:
  push:
    branches: [ main ]

jobs:
  performance-test:
    runs-on: ubuntu-latest
    steps:
      - name: Checkout repository
        uses: actions/checkout@v3

      - name: Set up Java 11
        uses: actions/setup-java@v3
        with:
          java-version: '11'
          distribution: 'temurin'

      - name: Download Apache JMeter
        run: |
          wget -q https://downloads.apache.org/jmeter/binaries/apache-jmeter-5.6.3.tgz
          tar -xzf apache-jmeter-5.6.3.tgz && mv apache-jmeter-5.6.3 jmeter

      - name: Run Performance Tests
        run: |
          mkdir -p results reports
          jmeter/bin/jmeter -n \
            -t tests/load_test.jmx \
            -l results/results.jtl \
            -e -o reports/

      - name: Upload HTML Report as Artifact
        uses: actions/upload-artifact@v3
        if: always()
        with:
          name: jmeter-report
          path: reports/
```

21. Commit and push all files to GitHub
22. Go to your repository on GitHub → click the Actions tab → watch the pipeline run automatically
23. After it completes → click the finished run → download the `jmeter-report` artifacts

Task 3 — Deliverables:

[]	Screenshot of the terminal showing JMeter running in non-GUI mode
[]	Screenshot of the HTML Dashboard index.html open in your browser
[]	Screenshot of the Statistics Table from the HTML dashboard
[]	Screenshot of the Response Time Over Time chart from the dashboard
[]	Written explanation (2–3 sentences): What does the P90 value in your report mean?
[]	GitHub repository link — share this with your instructor
[]	Screenshot of GitHub Actions pipeline completing successfully (green checkmark)
[]	Screenshot of the downloaded jmeter-report artifact from GitHub Actions

No local server needed — all tasks use free public APIs. Task 3 requires a free GitHub account. Each student's .jtl result file has unique timing values for plagiarism checking.

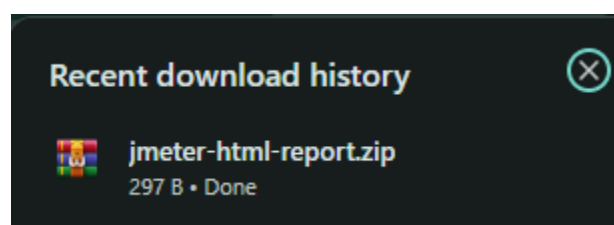
GIRHUB REPO LINK : <https://github.com/umair-fa22/AST-Terminal.git>



```

20 Artifact jmeter-html-report.zip successfully finalized. Artifact ID 7066359148
21 Artifact jmeter-html-report has been successfully uploaded! Final size is 297 bytes. Artifact ID is 7066359148
22 Artifact download URL: https://github.com/umair-fa22/AST-Terminal/actions/runs/2605339426/artifacts/7066359148

```



Submission Requirements box:

Submit one PDF report with screenshots and written answers, plus your .jmx test plan file. Include your Name, Student ID, and Date on the cover page.

Sample tables for Students [Will assist in VIVA]

Metric	Measured Value	Required Target	Pass / Fail	Notes
Avg Response Time	_____	_____	_____	_____
90th Percentile	_____	_____	_____	_____
Error Rate	_____	_____	_____	_____
Throughput	_____	_____	_____	_____

Results Table to Complete:

Test Run	Avg Response (ms)	Throughput (req/s)	Error Rate %	Max Response (ms)
1 User	_____	_____	_____	_____
10 Users	_____	_____	_____	_____
50 Users	_____	_____	_____	_____